

STUDYNEST CHROMA

Tailored Academic Reference & Study Portal for Amrita Students

Live Website: <https://krishnajith17.github.io/studynest/>

Presented by: Krishnajith (S1 EAC)

01. PROBLEM STATEMENT

- Tailored website for Amrita students for studies: Students need a dedicated platform customized specifically around the Amrita Vishwa Vidyapeetham B.Tech coursework.
- Course syllabi, reference textbook names, and unit-wise study materials are scattered across large PDFs, making daily study and quick revision difficult on mobile phones and laptops.

02. MOTIVATION

- One-stop destination for all the academic references for Amrita students: Bringing syllabus units, prescribed textbooks, reference books, and study notes together under a single link.
- Helping S1 and S2 students quickly check course topics, download unit PDFs, calculate SGPA, and use interactive study tools without searching multiple sources.

03. LIMITATIONS

- Being S1 students, we do not yet have a complete idea of Amrita examination question papers and the exact type of supplementary references or materials followed across all courses with respect to the Amrita coursework.
- Uploaded files in the Admin section are currently saved in browser local storage (up to 5 MB limit) rather than a centralized cloud server.

04. FUTURE IMPROVEMENTS

- We need to fine-tune the study materials, reduce redundancy, and add a more detailed, structured approach to learning the course materials.
- Include verified Amrita previous-year question papers (PYQs), faculty-approved lecture handouts, and cloud database storage so uploaded files sync for all students automatically.

05. WHAT THE WEBSITE DOES (KEY FEATURES)

- Complete Course Directory (11 Subjects): Covers Semester 1 & 2 B.Tech EAC courses (Calculus, IoT, Digital Electronics, Basic Electrical Engg, C Programming, Technical Communication, Cultural Education, etc.).
- Textbooks & Reference Books: Lists prescribed textbooks and reference books with author names and 1-click Google Books search links.
- Instant PDF Notes Download: Generates downloadable PDF study guides for the full syllabus or individual units in one click.
- Admin File Manager (Passcode: STUDY): Allows uploading new study files, removing old files, or exchanging (replacing) files on the website.
- SGPA Calculator & Learning Hub: Includes an Amrita credit-weighted SGPA calculator, Logic Gate simulator, and Ohm's Law / RC circuit calculator.
- Works on Mobile & Laptop + 5 Color Themes: Features a mobile bottom navigation bar and 5 vivid color themes with Dark/Light modes.

06. TOOLS & TECHNOLOGIES USED

- Frontend: React.js, Vite, HTML5, and Custom CSS3 (Responsive Mobile & Desktop Design).
- PDF & Storage: jsPDF for instant client-side PDF generation and Browser LocalStorage for saving custom files and bookmarks.
- Deployment: Hosted live on GitHub Pages with automated GitHub Actions deployment.

QUICK CLASSROOM DEMO STEPS

1. Open <https://krishnajith17.github.io/studynest/> and switch between the 5 Color Palettes.
2. Click any S1 course (e.g., 23ECE102 Digital Electronics) to view Units, Textbooks, and download a PDF.
3. Open SGPA Calculator & Learning Hub to show the grade estimator and logic gate simulator.
4. Click Admin (Passcode: STUDY) -> File Manager to Upload, Exchange, or Remove study files.